

Parent Functions II: x^3 and $\sqrt[3]{x}$

Algebra II | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph the cubic and cube root parent functions from a table and analyze their key attributes, including their rotational symmetry about the origin.

Vocabulary

- Rotational symmetry** — turning the graph _____ about the origin lands it back on itself.
- Reflectional symmetry** — the graph can be _____ across a line onto itself.
- Cube root** — $\sqrt[3]{a}$ asks: what number _____ gives a? Negatives _____ allowed, unlike square roots.

Part 1 — The Cubic Function, $f(x) = x^3$

Fill in the table. Notice what a negative input produces.

x	-2	-1	0	1	2
$f(x) = x^3$					

A negative input gives a _____ output, because a negative cubed stays negative.

Part 2 — The Cube Root Function, $f(x) = \sqrt[3]{x}$

This undoes the cubic. Compare it to your Part 1 table.

x	-8	-1	0	1	8
$f(x) = \sqrt[3]{x}$					

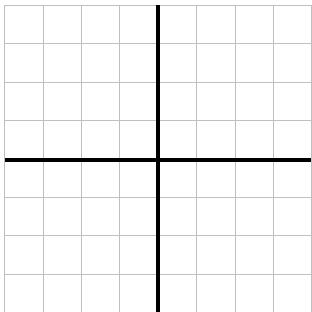
What do you notice next to Part 1? _____

We will come back to that on Day 10 - these two are _____.

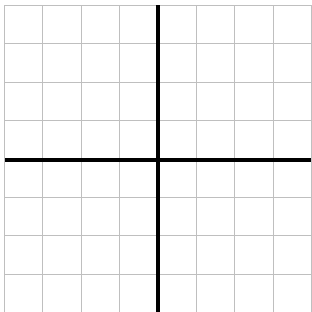
Part 3 — Graph Them Both

Each square is 1 unit. Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$ steep S-curve



B. $f(x) = \sqrt[3]{x}$ flatter S-curve



Passes through the _____

Mirror line between them: _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = x^3$	$f(x) = \sqrt[3]{x}$
Domain (interval)	_____	_____

Range (interval)	_____	_____
x- and y-intercept	_____	_____
Symmetry	_____	_____
Maximum or minimum?	_____	_____
Increasing / decreasing	_____	_____

These two share every attribute above. The difference is the _____ of the curve.

Part 5 — Compare to Day 4

1. Why does $\sqrt[3]{x}$ accept negatives when $\sqrt{x}$ does not? _____
2. Which of the four functions so far have a minimum? _____

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $(-3)^3 =$ _____ $\sqrt[3]{-27} =$ _____ $\sqrt[3]{64} =$ _____ $\sqrt[3]{0} =$ _____
2. Solve $x^3 = 125$. $x =$ _____ How many real answers? _____
3. Solve $x^2 = 25$. $x =$ _____ How many real answers? _____
4. Items 2 and 3 give different counts. Explain why cubing behaves differently from squaring.

5. $f(x) = x^3$ on the interval $[-1, 3]$: minimum _____ maximum _____
6. True or false: every real number has exactly one real cube root. _____
7. Which of the four parent functions so far accept negative inputs?

Exit Ticket

A function has domain and range both $(-\infty, \infty)$, passes through the origin, has 180° rotational symmetry, and no maximum or minimum.

Name two parent functions this could be: _____

What extra information would tell them apart? _____